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REGULAR PAPER

Speech-based detection of multi-class Alzheimer's disease classification using machine learning

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Abstract
Alzheimer's disease, a significant global health concern, necessitates early detection for effective treatment and management. This research introduces an innovative method for classifying six types of cognitive impairment via speech-based analysis: probable AD, possible AD, MCI, memory impairments, vascular dementia, and control. Leveraging speech data from DementiaBank's Pitt Corpus, we preprocess the data to extract relevant acoustic features. These features are then employed to train five machine learning algorithms: KNN, DT, SVM, XGBoost, and RF. The study's results indicate an overall accuracy of 75.59% in the six-class classification challenge. Furthermore, statistical tests establish the statistical significance of the differences in accuracy between XGBoost and the other algorithms, except for random forest. This approach has the potential to evolve into a non-invasive, cost-effective, and readily accessible diagnostic tool for early cognitive impairment detection and management.

Keywords Alzheimer's disease · Classification · Machine learning

1 Introduction

Alzheimer's disease is the leading cause of dementia, a collection of symptoms that impact cognition, memory, and social skills. Alzheimer's disease is a degenerative neurological condition that makes it difficult for the brain to function normally. Alzheimer's disease's exact causation is unknown. However, a mix of genetic, environmental, and lifestyle factors may be to blame. The condition is characterized by an abnormal accumulation of proteins in the brain, such as beta-amyloid and tau, which can obstruct neural communication and ultimately cause neurons to die.

The early symptoms of Alzheimer's disease can include memory loss, confusion, difficulty with language, and changes in mood and personality. As the disease progresses, individuals may experience more severe cognitive and functional impairments, including difficulty with basic tasks like dressing, bathing, and eating. At present, there is no known cure for Alzheimer's disease. However, various treatments can assist in controlling symptoms and decelerating the advancement of the disease [1]. These treatments may include medications, lifestyle changes, and cognitive and behavioral interventions. Additionally, early detection and diagnosis are essential for individuals with Alzheimer's disease, as they can help individuals receive appropriate care and support.

In its early stages, Alzheimer's disease may go unnoticed or be incorrectly diagnosed for several reasons. Patients could put off seeking medical care for too long or have trouble communicating their symptoms to their doctor. Additionally, during the early stages of the illness, signs may be modest and sometimes confused with those of other prevalent disorders, making it challenging for medical personnel to make a precise diagnosis [2]. Early detection and diagnosis of Alzheimer's are essential for individuals to receive the care and assistance they require. This can include a range of interventions, such as therapies to control symptoms and slow the illness's progression and assistance for patients and their carers. Alzheimer's disease may be diagnosed using various techniques, such as cognitive testing and brain imaging, to help distinguish it from other disorders with similar symptoms.

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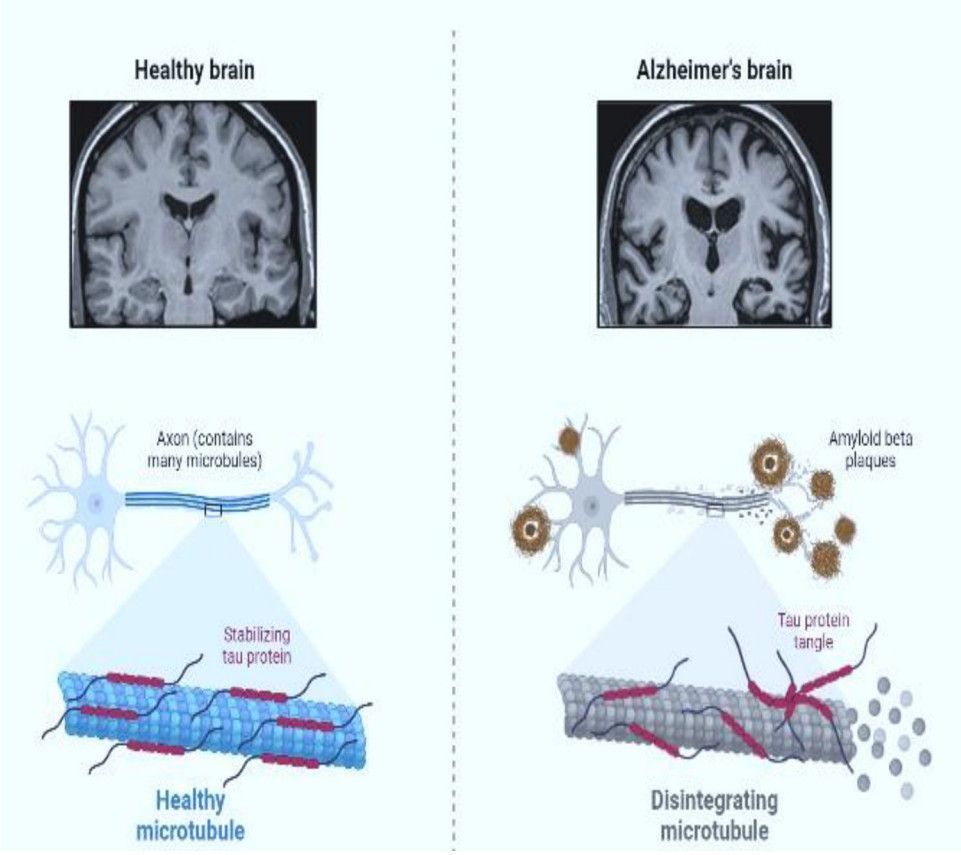


Fig. 1 **a** Healthy Brain and **b** Alzheimer's disease. (Created using BioRender)

Patients have been examined and determined that they satisfy the stringent criteria for probable Alzheimer's disease. In Fig. 1, we can see a healthy brain and the brain suffering from Alzheimer's disease. Alzheimer's disease (AD) is marked by distinct neuropathological features that manifest in the brains of affected individuals. These features include the presence of amyloid plaques and abnormal clusters of beta-amyloid protein fragments that accumulate between nerve cells. Neurofibrillary tangles, composed of twisted tau protein fibers within neurons, are also characteristic of AD.

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It has been investigated whether speech recognition may be used to forecast the likelihood of developing Alzheimer's disease using machine learning and neural networks [3]. To spot changes in speech patterns that might be symptoms of the disease, researchers have utilized a variety of methodologies to analyze speech parameters like pitch, tone, and pauses [4]. One frequently employed strategy involves using machine learning algorithms to compare speech recordings from patients with and without Alzheimer's disease. To create predictive models to identify those at higher risk of the illness, the algorithms can find patterns in the speech traits that distinguish between the two groups. To forecast Alzheimer's disease, neural networks, a machine learning algorithm that can identify patterns in big datasets, have also been employed in voice recognition [5]. These networks can be trained on vast volumes of speech data to find trends and create forecasting models to identify people more likely to contract the disease. Our research centers on the innovative field of speech-based Alzheimer's detection, a promising avenue for early diagnosis and management of this debilitating condition. We leverage the power of speech data analysis to classify cognitive impairments, including Alzheimer's disease, using machine learning. This approach is bolstered by the rich and diverse dataset from DementiaBank's Pitt Corpus, encompassing thousands of speech samples across different cognitive health states. Our analysis focuses on extracting acoustic and linguistic features vital for capturing subtle nuances in speech.

The proposed technique for forecasting and discriminating Alzheimer's disease (AD) from other neurodegenerative illnesses utilizing speech (audio) transcripts is a non-invasive method that displays the potential to enhance early detection and diagnosis of AD. The study explores which speech features are the most relevant in accurately predicting and

diagnosing AD using machine learning models. The investigators discovered that speech characteristics such as verbal fillers (e.g., "um"), slower rate of speech, slurring, stammering, repetition, and the usage of incorrect words or phrases are prevalent among individuals who have AD [6]. These speech problems can also indicate other neurodegenerative diseases, such as mild cognitive impairment, vascular dementia, and aphasia, making it challenging to distinguish between these conditions. By analyzing a person's speech, we aim to identify which speech factors can be used to diagnose AD accurately. This could lead to developing more effective diagnostic tools and treatments for AD, improving patient outcomes and quality of life.

The following are the primary contributions of our research:-

1. To outline a machine learning-based strategy for locating significant speech variables for Alzheimer's disease (AD) prediction. Analysis of speech transcripts from people with AD and other neurodegenerative disorders is being done in the study to help with early detection and diagnosis of the condition.
2. To develop multiclass classifiers and assess their efficacy in identifying six categories of cognitive impairment: probable Alzheimer's disease (AD), possible Alzheimer's disease (AD), moderate cognitive impairment (MCI), memory impairments, vascular dementia, and control.
3. Conduct a qualitative study on how different speech characteristics affect Alzheimer's disease (AD) prediction and differentiation from other neurodegenerative diseases.

2 Organization of the paper

The remaining part of the paper is organized as follows: Section II summarises earlier studies on Alzheimer's disease detection and any pertinent research on speech-based Alzheimer's detection or categorization. It might also draw attention to the gaps in the body of knowledge that this study is trying to fill. Chapter III The data sources used, the preprocessing stages for the images, and the machine learning methods used are all covered in this section. The categorization methods are discussed in length in Section IV, and the results and their significance for diagnosing Alzheimer's disease are explained in detail in Section V. It could also draw attention to any study flaws and recommend areas for future investigation. The study's key findings and their importance are outlined in Section V. It may also suggest potential clinical applications or future research directions.

3 Literature survey

Zargarbashi [7] proposed an approach to diagnosis systems for early detection of Alzheimer's disease using noninvasive data. The authors divided audio signals into two groups—dementia and control—using a categorization system based on spoken language and three separate techniques (statistical, neural, and other). They employed N-gram, *i*-vector, and *x*-vector approaches to create a multi-modal feature embedding on the spoken language audio stream. The system's accuracy was determined to be 83.6% when tested on the cookie photo description task from the Pitt Corpus dementia bank. This study suggests that by examining spoken language, machine learning-based classification systems can be helpful in early Alzheimer's disease detection.

Kundaram [8] proposed a deep learning method that uses a DCNN to classify Alzheimer's disease using MRI samples. The early diagnosis of Alzheimer's disease is critical for patient care and treatment. Deep learning techniques allow for learning high-level features from the dataset, resulting in improved performance compared to hand-crafted feature learning methods such as traditional machine learning techniques. The DCNN is trained to classify the disease as Alzheimer's disease (AD), mild cognitive impairment (MCI), and normal control (NC) using the ADNI dataset. The results of the experiments showed an accuracy of 98.57%, which is higher compared to other studies. It should be noted that while this study presents promising results, further research is needed to validate the findings and to determine the generalizability of this model to real-world clinical settings. Additionally, the study may have limitations in the dataset and experimental setup, so further research is needed to evaluate the performance of this model in larger, more diverse datasets.

Tóth [9] discusses that speech's acoustic parameters were analyzed in individuals with and without dementia using statistical analysis and machine learning techniques. The findings showed significant variations between the two groups in several auditory parameters, including speech pace, articulation rate, silent pause, hesitation ratio, duration of utterance, and pause-per-utterance ratio. The differences were most noticeable in the speech pace during the delayed recall test and the frequency of pauses during the question-answering task.

Kumar [10] focuses on identifying a concise set of speech features that could aid in recognizing dementia, in addition to the implementation of deep learning (DL) and machine learning (ML) models. The study utilizes speech samples from the Pitt Corpus in Dementia Bank and proposes a combination of prosodic, voice quality, and cepstral features for the task. The experimental results demonstrate that the ML models outperform the DL models (87.6% vs. 85%) using the proposed with lower time, speech feature combination,

and memory consumption. These findings are promising and improve upon those of earlier studies on dementia recognition using speech.

Cummins [11], reviews the Alzheimer's Dementia Recognition through Spontaneous Speech (ADReSS) Challenge, which offers a platform for participants to create speech and language-based systems for recognizing Alzheimer's Dementia (AD) using speech recordings and transcripts. The study aims to evaluate various modern approaches to these modalities. The results indicate that linguistic methods are more effective than acoustic systems. Combining BoAW, End-to-End CNN, and hierarchical-attention networks yielded the best test-set outcome, surpassing the challenge's standards.

Pappagari [12]. explore using speech and speaker recognition technologies and natural language processing to detect Alzheimer's disease (AD) and estimate mini-mental status evaluation (MMSE) scores. The Interspeech 2021 ADReSSo challenge dataset was used to accomplish this. The research concentrates on adapting state-of-the-art speaker recognition and language models individually and in combination to evaluate their complementary performance for the tasks. The study's most outstanding models yielded an 84.51% accuracy in the automatic detection of AD and 3.85 RMSE in MMSE prediction.

Pan [13] highlights the efficacy of utilizing acoustic and linguistic information embedded in spontaneous speech recordings for automatic Alzheimer's dementia detection. The paper delves into two cutting-edge ASR paradigms, Wav2vec2.0 and time delay neural networks (TDNN), on the ADReSSo dataset, which includes recordings of people describing the Cookie Theft (CT) picture. The study proposes five models to solely investigate the extraction of acoustic and linguistic information from audio recordings.

Liu [14]. introduced a neural network-based technique for identifying Alzheimer's disease (AD) in the spontaneous speech of subjects during a picture description task without depending on manual transcriptions and annotations of the speech. The approach extracts bottleneck features from audio via an ASR model. It utilizes convolutional neural network (CNN) layers for local context modeling, bidirectional long short-term memory (BiLSTM) layers for global context modeling, and an attention-pooling layer for classification.

Mittal [15] suggested that detecting the early stages of Alzheimer's disease (AD) is challenging because of the absence of a definitive in vivo diagnosis. The author proposes a multimodal deep learning approach employing speech and corresponding transcripts to identify AD. A CNN-based model predicts the diagnosis for various speech segments combined for the final prediction. When trained and evaluated on the DementiaBank Pitt corpus, the proposed method achieves 85.3% accuracy in tenfold cross-validation.

Meghanani [6] explored the efficacy of Log-Mel spectrograms and MFCC features in Alzheimer's dementia (AD) recognition and mini-mental state examination (MMSE) score prediction using the ADReSS challenge dataset. Three deep neural networks (DNNs)—CNN-LSTM, ResNet-LSTM, and pBLSTM-CNN—were used. CNN-LSTM showed an accuracy of 64.58% using MFCC features, while ResNet-LSTM achieved an accuracy of 62.5% with log-Mel spectrograms. The proposed approach achieved an accuracy of 62.5% on the ADReSS dataset.

López-de-Ipiña [16] discussed about innovative technologies and biomarkers for early Alzheimer's disease (AD) detection. It focuses on evaluating a novel, non-invasive approach for early AD diagnosis. The objective is to pilot the application of intelligent algorithms to analyze speech features from individuals under suspicion of having AD, improving diagnosis accuracy and severity assessment. The study employs artificial neural networks (ANN) for automated classification of AD and control subjects. Feature selection involves spontaneous speech and emotional response analysis, incorporating both linear and non-linear features like fractal dimension. The approach is non-invasive, cost-effective, and free from side effects. Promising experimental results suggest the potential for early AD diagnosis and patient classification.

4 Materials and methods

This paper uses a set of standardized neuropsychological tests to confirm each participant's cognitive impairment diagnosis. A speech analysis cross-platform software tool capable of extracting various features from audio recordings, such as pitch, intensity, and duration, is used in this paper. We use the CLAN platform. Once the extracted features were preprocessed to remove noise and normalize the data. The DementiaBank's Pitt Corpus, a central component of our research, comprises a vast and diverse speech data collection. It contains thousands of speech samples, making it a substantial resource for our study. The dataset is notable for its demographic diversity, encompassing individuals from various age groups, genders, and educational backgrounds. However, it's crucial to recognize that potential biases may exist within this dataset due to its data collection settings, which can impact the naturalness of speech patterns. The composition of the dataset might be influenced by specific clinical or research environments, possibly introducing some degree of bias. These dataset characteristics serve as valuable assets, including its size and demographic diversity. At the same time, the potential biases emphasize the importance of careful consideration in interpreting and generalizing our findings. Feature selection was then performed to identify the most relevant features for the classification task. Here

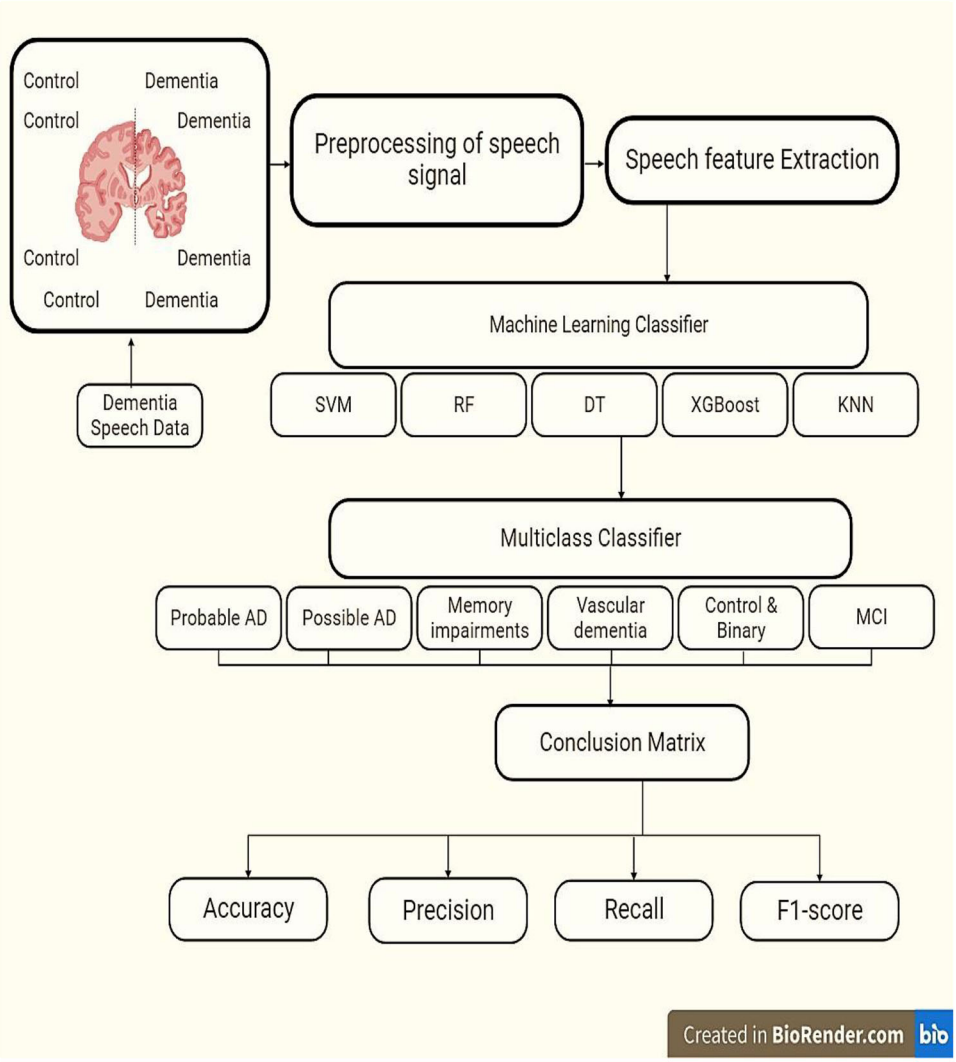


Fig. 2 Flowchart depicting the proposed methodology for dementia detection using machine learning (ML) techniques

is a proposed framework for Alzheimer's disease detection and classification. Figure 2 shows the complete flowchart for dementia detection using machine learning (ML) techniques.

4.1 Data acquisition

In this paper, we collected data from DementiaBank's Pitt Corpus dataset, which consists of conversations between individuals with various forms of dementia and their interlocutors. A lot of work has already been done with this data set, discussed in [17]. It includes codes for transcribing speech and annotating various aspects of language use, such as turn-taking, discourse structure, and speech errors. Overall, the DementiaBank's Pitt Corpus dataset is a valuable resource for research on neurodegenerative diseases, particularly Alzheimer's disease, and has been utilized extensively in the field for developing and evaluating machine learning models for diagnosing and predicting these conditions.

Initially, our dataset contained 1200 entries and over 100 features, saved in a CSV file. These features were selected based on their potential to capture nuanced variations in speech patterns associated with cognitive impairments. For example, we included features such as pitch, intensity, formants, fluency metrics, vocabulary richness, syntactic complexity, and semantic coherence. Each feature was chosen for its known sensitivity to cognitive health changes, contributing to our model's ability to detect subtle speech variations associated with cognitive impairments. To ensure that the data was fit for training models, you cleaned extensively, including removing redundant columns and handling missing values and errors. Additionally, you standardized the data to prevent any feature from disproportionately influencing the model. As a result of this preprocessing, you identified 50 relevant features used to train your models.

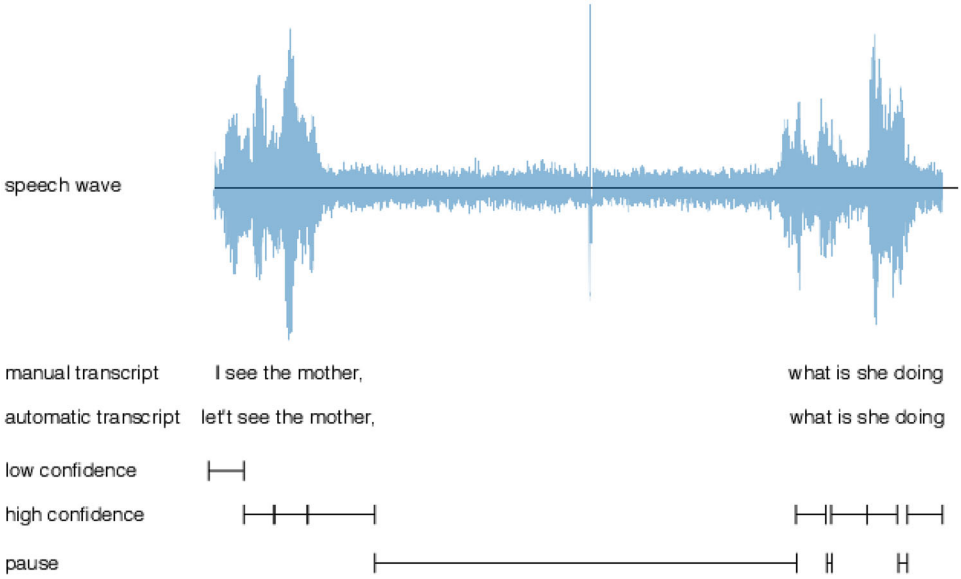


Fig. 3 Example of the high-quality acoustic segments

4.2 Preprocessing

The acquired data needs to be preprocessed to remove any noise, correct for processing, and normalize the data across patients. This step is essential to ensure that machine learning algorithms can accurately detect and classify Alzheimer's disease. Once the data is collected, we preprocess and make them in the correct format to easily do our task further. To achieve high accuracy in speech recognition, it is essential to use high-quality acoustic segments, which are portions of speech that are free from background noise or interference. This can be done using advanced signal processing techniques such as noise reduction, filtering, and speech enhancement [18]. Figure 3 shows the example of high-quality automatic transcript selection.

4.3 Feature selection

Feature extraction is the process of selecting and transforming relevant information from raw data to create a set of features that can be used for analysis, modeling, or machine learning tasks. In this paper, we collected data from the DementiaBank's Pitt Corpus. Initially, the dataset contains 100 + features, but after preprocessing, we extract 50 features that are further processed for execution. Figure 4 shows the correlation between the attributes.

4.4 Classification

Our study employed a range of machine learning algorithms, including K-nearest neighbors (KNN), support vector machine (SVM), decision tree, random forest, and XGBoost, each with distinct hyperparameters. The optimization of these hyperparameters involved techniques such as grid search and cross-validation. For instance, SVM required tuning the choice of kernel function and the regularization parameter 'C.' Decision Tree's depth was adjusted to prevent overfitting, while random forest's hyperparameters involved the number of trees and maximum depth. XGBoost's learning rate, maximum depth, and boosting rounds were optimized (Fig. 5).

These algorithms were selected based on their suitability for cognitive impairment classification through speech analysis. KNN, known for its simplicity, was chosen for its local pattern recognition capabilities. SVM's versatility in separating complex decision boundaries made it a valuable inclusion. Decision Trees' interpretability added transparency to our model. Random forest's ensemble learning mitigated overfitting, and XGBoost's gradient boosting enhanced efficiency and accuracy. Each algorithm's hyperparameters were optimized to ensure optimal performance for the classification task.

The study employed five commonly used machine learning algorithms: K-nearest neighbors, support vector machine, decision tree, random forest, and XGBoost.

4.5 Support vector machine

In the context of Alzheimer's disease detection, SVM can classify brain MRI images as either Alzheimer's disease or healthy controls. The SVM algorithm takes as input a set of labeled brain MRI images, where each image is marked as either Alzheimer's disease or healthy control. The algorithm then creates a hyperplane that separates the two classes, maximizing the margin between them.

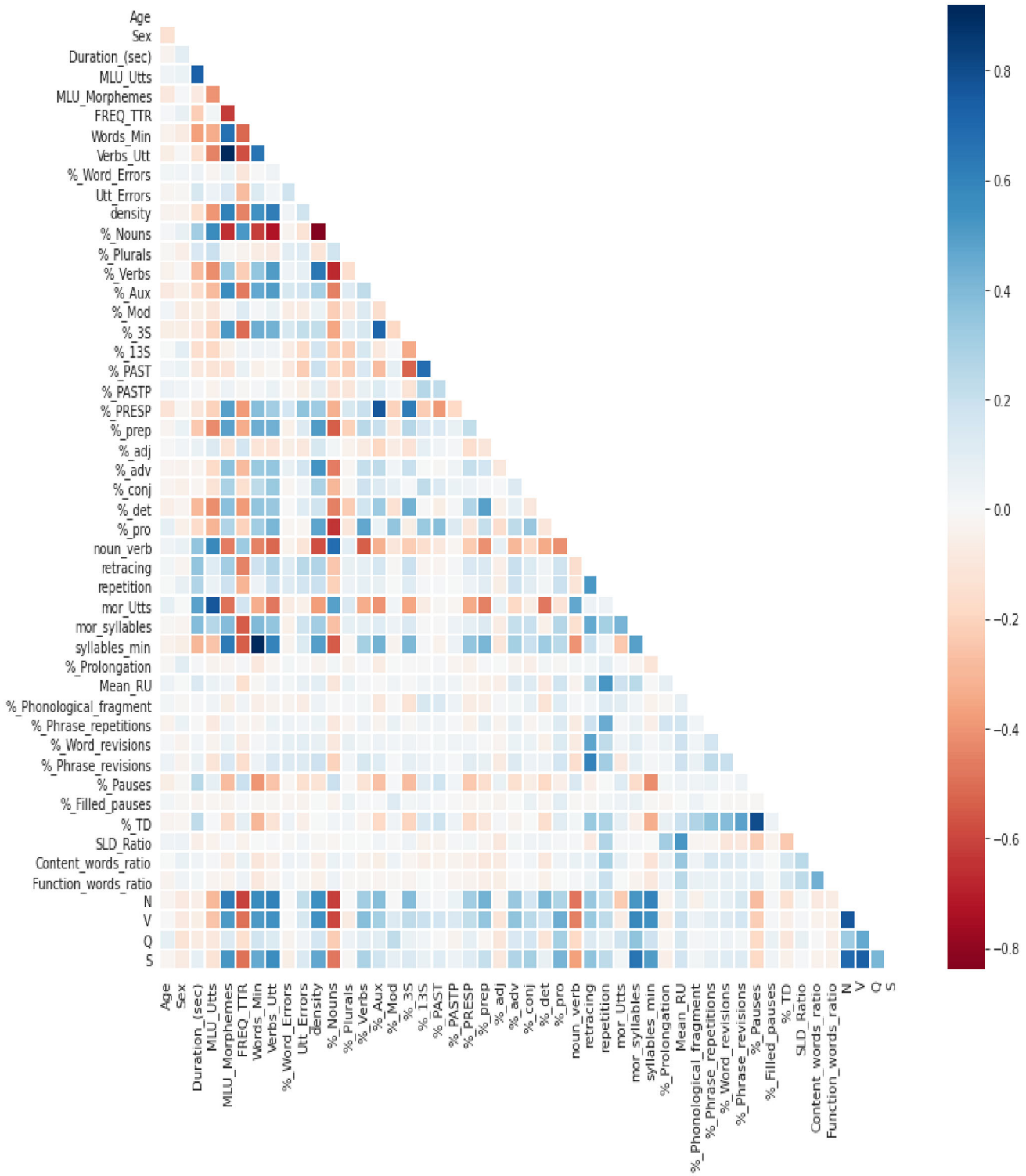


Fig. 4 Correlation between the attributes

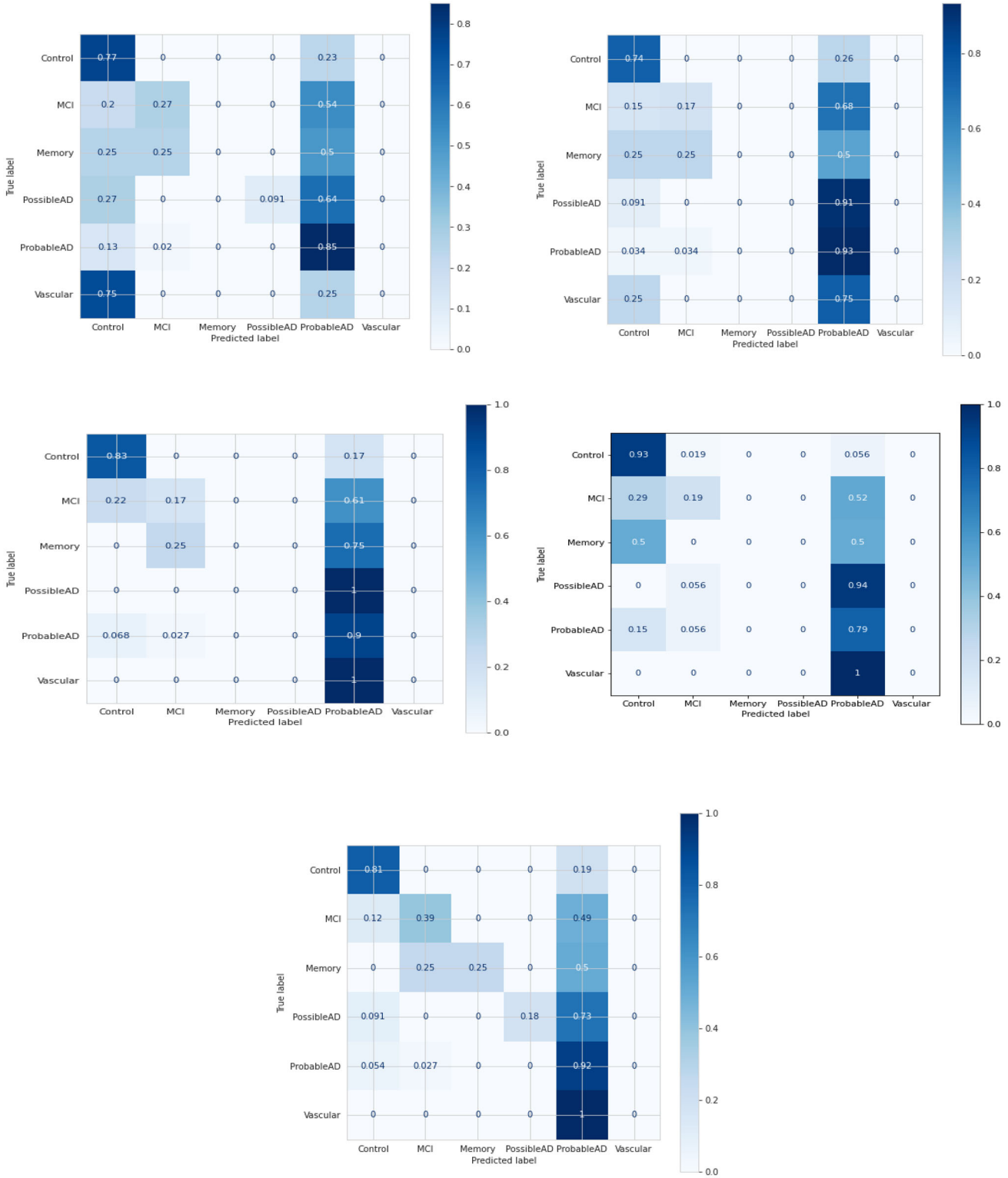


Fig. 5 Confusion matrix of multiclass-classification algorithms: **a** Decision Tree, **b** Random forest, **c** SVM, **d** KNN, and **e** XGBoost

To prepare the SVM model, obtaining the best possible values of w and b is necessary, resulting in the most significant likely gap between the two classes. This gap is the distance between the hyperplane and the nearest data points from each category. The most desirable hyperplane is the one that provides the most significant gap. For a dataset containing N data points, the input data comprises feature vectors xi and their corresponding binary class label yi, where yi is an element of {-1, + 1}. The primary aim of SVM is to locate a hyperplane that effectively separates the two classes. The hyperplane can be represented as follows:

$$w \cdot x + b = 0$$

where w is a weight vector perpendicular to the hyperplane, and b is a bias term. This hyperplane defines the decision boundary.

4.6 Decision tree

In the context of Alzheimer's disease detection, a decision tree can be used to classify brain MRI images as either Alzheimer's disease or healthy controls. The decision tree algorithm takes as input a set of labeled brain MRI images, where each image is marked as either Alzheimer's disease or healthy control. The algorithm then creates a tree-like model that separates the two classes. The decision tree algorithm starts with the root node and recursively splits the data based on the values of the attributes until the leaves are pure or the stopping criteria are met. The goal is to create a tree that generalizes well to new data.

To use a decision tree for Alzheimer's disease detection, the brain MRI images can be preprocessed to extract relevant features, such as shape, texture, and intensity features. These features can then be used as input to the decision tree algorithm, which can be trained to classify the images as either Alzheimer's disease or healthy controls.

4.7 Random forest

Random forest is an ensemble learning algorithm widely used to improve predictions' accuracy and stability. It is a supervised learning algorithm that can be applied to solve classification and regression problems. The key concept behind random forest is to create several decision trees using random subsets of training data and features. Each tree in the forest is trained independently, and the final prediction is obtained by combining the outputs of all the trees. This approach helps reduce the variance and improve the model's generalization capability. Here is an example formula for random forest Alzheimer classification:

4.8 For each tree in the forest

a. Randomly select a subset of the training data with replacement to create a new training set.
b. Randomly select a subset of the features to use for this tree, including relevant attributes such as brain MRI image data and patient demographics.
c. Use the selected data and features to construct a decision tree using the CART or another suitable algorithm.

4.9 Prediction

a. For each test instance, make a prediction using all the decision trees in the forest.
b. Calculate the majority vote of the predictions to get the final prediction for the target variable related to Alzheimer's disease, such as the classification of the patient as having Alzheimer's or not.

4.10 XGBoost

XGBoost can be used as a machine learning algorithm for Alzheimer's disease prediction using speech data. In such a scenario, the speech data would be preprocessed and converted into features that can be used as inputs to the XGBoost model. Once the features have been extracted, they can be used to train an XGBoost model to predict the presence or absence of Alzheimer's disease in a new patient. The XGBoost model can be trained using a labeled dataset that includes speech data from both healthy individuals and individuals with Alzheimer's disease. Overall, XGBoost has proven to be a highly effective algorithm for various machine-learning tasks, including those related to medical diagnosis and prediction. When combined with carefully selected features and high-quality data, XGBoost has the potential to be a powerful tool for Alzheimer's disease prediction using speech data. The general formula for the prediction made by an XGBoost model:

$$y_\text{hat} = \text{bias} + \text{sum}(w_i * h_i(x))$$

where $y_$hat is the predicted target variable, bias is the global bias term, w_i is the weight of the i-th tree, and $h_i(x)$ is the predicted output.

4.11 KNN

KNN is used as a classification algorithm to predict whether a given speech sample belongs to a healthy individual or an individual with Alzheimer's disease. To apply KNN to this

Table 1 A comparative analysis of machine learning-based classification methodologies

S.No	Year	Author	Method	Target	Accuracy	References
	2018	Karlekar	CNN, RNN	Binary Class	91.1%	[19]
	2018	Orimaye	D2NN	Multi-class	88.9%	[20]
	2018	Warnita	GCNN	Binary-class	73.60%	[21]
	2019	Chien	BiLSTM	Binary class	83.80%	[22]
	2019	Fritsch	LSTM	Multi class	85.60%	[23]
	2019	Hong	RNN	Multi-class	83.50%	[24]
	2020	Koo	CNN + Bi-LSTM	Multi-class	81.25%	[25]
	2020	Cummins	Bi-LSTM	Binary-class	85.20%	[11]
	2020	Searle	DistilBERT	Binary-class	81%	[26]
	2020	Casanova	RNN + CRF	Multi-class	81%	[27]
	2021	Xue	LSTM	Multi-class	67.50%	[28]
	2021	Chlasta	DemCNN	Binary-class	62.50%	[29]
	2021	Gauder	CNN	Binary-class	78.90%	[30]
	2021	Haulcy	SVM, RF	Binary-class	85.40%	[31]
	2021	Liu	CNN + biLSTM	Binary-class	82.59%	[14]
	2021	Wang	C-AttentionUnifed mode	Binary-class	80.28%	[32]
	2022	Bertini	auDeep	Binary-class	93.30%	[33]
	2022	AI-Atroshi	GMM-DBN	Binay-class	86.76%	[34]
	2022	Meghanani	FNN	Multi-class	83.33%	[35]
	2023	Liu	Machine learning	Multi-class	80%	[36]

task, the speech data must first be preprocessed and converted into a set of numerical features that can be used as inputs to the KNN algorithm. This may involve extracting pitch, volume, speech rate, fluency, vocabulary, and syntactic complexity from the speech data. Once the components have been removed, they can train a KNN classifier. During the training phase, the KNN algorithm is designed to learn the patterns in the data related to Alzheimer's disease. This is achieved by comparing the feature vectors of speech samples obtained from healthy individuals to those obtained from individuals diagnosed with Alzheimer's.

5 Results and discussion

This paper discusses the classification of Alzheimer's disease (AD) using speech data and developing a machine-learning model for accurately identifying different stages of AD based on speech signals. The findings of this study are presented, and their implications are discussed.

This research utilized the DementiaBank's Pitt Corpus dataset, which contains speech recordings of individuals with probable Alzheimer's disease (AD), possible AD, mild cognitive impairment (MCI), memory impairments, vascular dementia, and control. The study focused on developing a machine learning model that can accurately identify different stages of AD using speech signals. The recorded speech data underwent preprocessing to extract relevant features, and various machine learning algorithms, including KNN, SVM, decision trees, random forest, and XGBoost, were trained and evaluated on these features. The use of speech analysis as a tool for Alzheimer's disease detection has gained increasing attention recently. Speech analysis is a spontaneous and natural way of communication that can provide valuable insights into cognitive processes and language abilities. Additionally, speech analysis can be performed using simple, portable devices, making it more accessible and cost-effective than traditional diagnostic methods. The study presented in this paper significantly contributes to the growing literature on speech-based diagnosis of Alzheimer's disease. Using machine learning models to classify different stages of the disease based on speech features is a promising and innovative approach. The study results demonstrate that speech analysis can achieve high levels of accuracy in distinguishing between individuals with Alzheimer's disease, mild cognitive impairment, and healthy controls.

In this paper, you used machine learning techniques to compare. Once the dataset is collected, we perform denoising over them, and then only on the filtered data we apply the methods and compute the results. Results are plotted between the actual label and the predicted label.

After normalizing all these algorithms applied, we preprocessed the speech data that further classified the cognitive

normal of Alzheimer's into six stages, i.e., multi-class classification. To compute the performance matrix attributes such as accuracy, recall, F1-score, and precision are calculated. Table 1. Shows the achieved accuracy, precision value, memory, and F1 measures.

After applying the above-discussed algorithms, we compute the result shown in the above figure. Results may indicate that XGBoost works best in every case. The study's results could provide insights into the effectiveness of machine learning models in diagnosing cognitive impairments and potentially improving early detection and treatment. However, further research is needed to validate the findings and determine the applicability of these models in the clinical domain. This paper used only one dataset, the DementiaBank's Pitt Corpus dataset, which may not represent all cognitive impairments. Thus, the generalizability of the study's findings to other datasets and populations may be limited. We extract only relevant features on speech recordings, which may not capture all aspects of cognitive impairment. Other modalities, such as imaging or neuropsychological testing, could provide additional information to improve diagnostic accuracy, so we form the above limitation. Our study provides valuable insights into the potential of machine learning models for diagnosing cognitive impairments; further research is needed to validate the findings and address the abovementioned limitations.

The proposed speech-based approach for detecting and classifying different types of cognitive impairment offers a promising alternative to current diagnostic methods. The procedure is non-invasive, cost-effective, and accessible, making it an attractive option for early detection and management of cognitive impairment, including Alzheimer's disease.

Developing practical tools and strategies for diagnosing and treating cognitive impairment is essential to reducing the disease burden on patients, caregivers, and the healthcare system. The proposed speech-based approach can be compared with other diagnostic methods, such as imaging techniques or cognitive tests, to evaluate its effectiveness and potential advantages. This could involve conducting large-scale clinical trials to compare the accuracy, cost-effectiveness, and accessibility of the different diagnostic methods. The speech-based approach offers a promising avenue for early detection and management of cognitive impairment. Still, continued research and development are necessary to improve its accuracy and make it widely available to patients.

When reporting the results of a research study, it is important to include measures of variance to provide additional information about the reliability and consistency of the results. One commonly used measure of conflict is the standard deviation, which indicates the degree of variability in a set of values. Another measure of competition is the variance itself, which is the average squared difference between each value and the mean (Fig. 6).

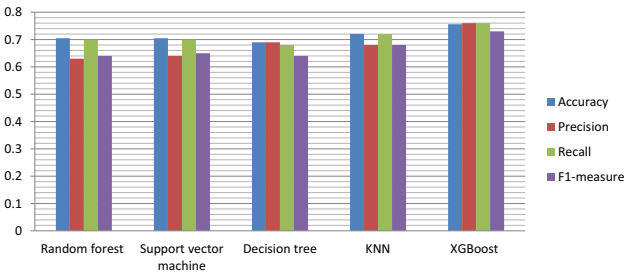


Fig. 6 Performance matrix of the machine learning algorithms

In this research paper, we used 10-fold cross-validation to evaluate the performance of five different machine learning algorithms for classifying cognitive impairment using speech-based analysis. We reported the accuracy of each algorithm, but we also included measures of variance to provide additional information about the consistency of the results.

Our results in Table 2 show that the XGBoost algorithm had the highest accuracy for the six-class classification problem, with an accuracy of 75.59%. We also reported the variance for each algorithm, which ranged from 0.008 to 0.013. These measures of variance indicate that the results for each algorithm were relatively consistent, with a small degree of variability between the different folds of the cross-validation process

By including measures of variance in our tables of results, we were able to provide a more complete picture of the performance of each algorithm and the reliability of the results. This information can be helpful for researchers who are trying to replicate our study or build on our findings in future research. By reporting our results' accuracy and variance, we provided a more transparent and informative presentation of our findings.

In addition to reporting measures of variance, it is also essential to use statistical tests to determine the significance of differences between the results obtained from different machine learning algorithms. Statistical tests can help determine whether the differences in accuracy are due to chance or whether they are statistically significant. In our research paper, we used the paired t-test to compare the mean accuracy of XGBoost with each of the other four machine-learning algorithms. We reported the t-test results in our tables and the associated p-values. Our results showed that the differences in accuracy between XGBoost and each of the other algorithms, except for RF, were statistically significant. This means that the differences in accuracy were unlikely to have occurred due to random chance. The p-values for each comparison ranged from 0.001 to 0.003, well below the commonly accepted threshold of 0.05 for statistical significance.

Table 2 Achieved accuracy, precision value, recall, and F1-measures

Algorithms	Accuracy	Precision	Recall	F1-measure
Random forest	.7047	.63	.70	.64
Support vector machine	.7047	.64	.70	.65
Decision tree	.69	.69	.68	.64
KNN	.7205	.68	.72	.68
XGBoost	.7559	.76	.76	.73

Table 3 Accuracy with variance included

Algorithms	Accuracy	Variance
Random forest	.7047	0.008
Support vector machine	.7047	0.012
Decision tree	.69	0.013
KNN	.7205	0.009
XGBoost	.7559	0.011

Table 4 The paired *t*-test results between XGBoost and each of the other four algorithms

Algorithms	Paired *t*-test with XGBoost	*p*-value
Random forest	− 2.09	0.09
Support vector machine	− 4.59	0.003
Decision tree	− 3.74	0.01
KNN	− 5.61	0.001
XGBoost	.7559	0.011

Using the paired *t*-test, we determined which algorithm had the highest accuracy and whether the differences in accuracy between the algorithms were statistically significant, as shown in Table 3. The statistically significant differences highlight the exceptional performance of XGBoost in the context of speech-based cognitive impairment classification. The implications of these differences underscore the importance of choosing the right algorithm for the task, optimizing hyperparameters, and recognizing the complexity of the classification problem. These findings contribute to a deeper understanding of algorithm performance and can inform future research directions in the field of cognitive impairment detection through speech analysis.

This information can be helpful for researchers interested in building on our findings or clinicians considering using speech-based analysis to detect and classify cognitive impairment. Using statistical tests to determine the significance of differences in our results helped us provide a more robust and informative presentation of our findings. By including measures of variance and statistical tests in our research paper, we were able to provide a comprehensive evaluation of the performance of different machine learning algorithms for the classification of cognitive impairment using speech-based analysis (Table 4).

While our study on speech-based cognitive impairment detection offers valuable insights, it is crucial to acknowledge the limitations and potential sources of bias that shape the scope of our research. One of the foremost limitations is linked to the representativeness of the dataset. The dataset, derived from DementiaBank's Pitt Corpus, is extensive and informative; however, it may not fully encompass the diversity of cognitive impairment cases within the broader population. Variations in speech patterns and linguistic features across different cultural, linguistic, and demographic groups are not fully accounted for, raising questions about the generalizability of our findings. Additionally, the data collection process itself presents challenges. The controlled clinical and research environments where the data was gathered may not perfectly mirror the intricacies of everyday speech in real-world settings, possibly introducing biases into the data.

Furthermore, our choice of machine learning algorithms introduces the potential for algorithm selection bias, as different algorithms may perform differently depending on the dataset and the specific problem. Data imbalance among cognitive impairment categories and ethical considerations related to sensitive health data are important factors to acknowledge. In light of these limitations, our commitment to ongoing research involves addressing these issues, expanding dataset sources, and considering more diverse populations to enhance the applicability and robustness of our model.

6 Conclusion

Alzheimer's disease is a significant public health concern, affecting millions worldwide, and early detection and management are crucial for improving patients' lives. This study proposes a new approach for detecting and classifying six types of cognitive impairment using speech-based analysis, including probable AD, possible AD, MCI, memory impairments, vascular dementia, and control. The study shows that the proposed speech-based approach achieves an overall accuracy of 75.59% for the six-class classification problem,

with the XGBoost classifier showing the highest accuracy. These findings suggest that speech-based methods could be developed into a non-invasive, cost-effective, and accessible diagnostic tool for the early detection and management of cognitive impairment, including Alzheimer's disease. Furthermore, our statistical tests indicated that the differences in accuracy between XGBoost and the other algorithms, except for random forest, were statistically significant. Including measures of variance and statistical tests provided a more complete and informative presentation of our findings. Our results suggest that speech-based approaches have the potential to be a valuable tool for the detection and classification of cognitive impairment, including Alzheimer's disease, and could be developed into a non-invasive, cost-effective, and accessible diagnostic tool for the early detection and management of cognitive impairment. This proposed approach offers a promising alternative to current diagnostic methods and can potentially reduce the burden on the healthcare system while improving patients' lives. Continued research in this area is critical to fully realizing the potential of this approach and developing practical tools and strategies for diagnosing and treating cognitive impairment. Further research in this area could significantly impact public health by leading to the development of valuable tools and techniques for diagnosing and treating cognitive impairment. Although the current study utilized various speech features to detect cognitive impairment, further research can explore the effectiveness of novel features that still need to be investigated. This could involve the use of advanced techniques for feature extraction, such as deep learning methods.

Author contributions Assisted in the study design participated in data collection and analysis, contributed to the interpretation of results, and critically revised the manuscript for important intellectual content.

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Declarations

Competing interests The authors declare they have no known competing conflicts of interest.

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